Gaming the Two Numbers They Control: Quantization, Weight-Cut Bunching, and the Limits of Strength in 3.9 Million Powerlifting Results

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***Abstract*—A powerlifting competitor controls exactly two numbers: the weight loaded on the bar and their bodyweight at weigh-in. We ask whether these choices leave measurable statistical fingerprints, using $\sim$3.94 million competition results from the public-domain OpenPowerlifting database. Four findings, each answered with a course-appropriate test and led by effect size (at $n \approx 3.9$M almost everything is "significant"). (i) *Quantization*: 96.2% of 13.4M attempt loads sit exactly on the 2.5 kg plate grid; a generalized likelihood-ratio test (GLRT) whose null parameter lies on the boundary is calibrated by a parametric bootstrap. (ii) *Weight-cut bunching*: bodyweight piles up just below every IPF class limit (de-heaped $\log(\text{below}/\text{above}) = +1.92$ at 83 kg); a Cattaneo–Jansson–Ma density-discontinuity test rejects at all seven limits (surviving Holm and Benjamini–Hochberg), a non-limit control shows no bunching, and placebos show no false positives. (iii) *Allometry*: strength scales with bodyweight differently by sex ($b = 0.75$ men, 0.51 women vs. the isometric $2/3$; a Sex$\times \log$(BW) interaction is highly significant and persists on common support). (iv) *Prediction*: basic controllable variables predict the total ($R^2 = 0.70$, random forest, leakage-safe grouped CV). The strategy leaves a measurable fingerprint. Code and data: https://github.com/adirelm/opl-stat-theory.**

I. Introduction

Powerlifting comprises three lifts—squat, bench press, and deadlift—each with three attempts, contested within bodyweight classes. Two features make the sport an unusually clean laboratory for strategic behaviour. First, the implements are *discrete*: plates come in fixed denominations, so the load a competitor can place on the bar is not a continuous choice. Second, eligibility is governed by a *threshold*: a lifter who weighs even slightly above a class limit is bumped into a heavier, stronger field. A competitor's agency therefore reduces to two numbers—the load chosen for each attempt, and the bodyweight registered at weigh-in—and both interact with a sharp administrative rule. We ask whether these two choices leave detectable statistical signatures, and what $\sim$3.94M results reveal about the limits of human strength.

Related work. Manipulation of a "running variable" around an administrative threshold produces *bunching*, a workhorse idea in public economics [1] that is classically tested with the McCrary density-discontinuity test [2] and its modern, bias-corrected local-polynomial form due to Cattaneo, Jansson, and Ma [3]. Rapid pre-weigh-in weight cutting is well documented in weight-class sports. On the physiological side, the square–cube law predicts that strength, governed by muscle cross-sectional area, scales with bodyweight as $\propto \mathrm{BW}^{2/3}$. Extreme-value theory has been used to estimate the ultimate limits of athletic performance from record data [6]. On OpenPowerlifting specifically, recent work studies *scoring and scaling* systems [7]; to our knowledge, no formal manipulation test of weight-cut bunching has been applied to this dataset.

Contribution. We tell one story—"the two numbers and the limits of strength"—that deliberately spans both halves of the course, classical inference and machine learning. Our contributions are: (1) a rounded-likelihood *quantization* GLRT for the plate grid, with a parametric-bootstrap calibration that respects the boundary nature of the null; (2) to our knowledge the first formal density-discontinuity test of weight-cut *bunching* on OpenPowerlifting, hardened with a non-limit control, placebo cutoffs, de-heaping, and multiple-testing corrections; (3) a sex-interaction *allometry* model that tests the scaling exponent against the isometric $2/3$; and (4) a leakage-safe strength-*prediction* model. Throughout we lead with effect sizes and confidence intervals rather than p-values, for reasons we now make explicit.

Methodological stance. At $n \approx 3.9$M the power to detect any non-zero effect is effectively one, so a p-value answers a question we do not care about ("is the effect exactly zero?") rather than the one we do ("is it large?"). We therefore report effect sizes with 95% confidence intervals as the primary evidence, use p-values only as a secondary screen, and apply all four standard multiple-testing corrections—Bonferroni, Holm, Šidák (familywise), and Benjamini–Hochberg (false-discovery-rate)—to every pre-declared family of tests. Because the same lifter appears in many rows, naive standard errors understate uncertainty (pseudo-replication); the formal tests therefore use one row per lifter or cluster-robust standard errors, declared per hypothesis.

II. Results

All intervals are 95%. Headline effect sizes use all rows; formal tests use one row per lifter (to respect independence) and, where the class scheme matters, the modern kg-only era (2014+).

A. Attempt loads are quantized to the 2.5 kg grid (H1)

Of 13,397,702 valid attempt loads, **96.20%** (CI $[96.19, 96.21]$) fall exactly on the 2.5 kg plate grid (Fig. 1); the smallest standard plate is 1.25 kg, loaded on both sides, so 2.5 kg is the minimum increment. To test this formally

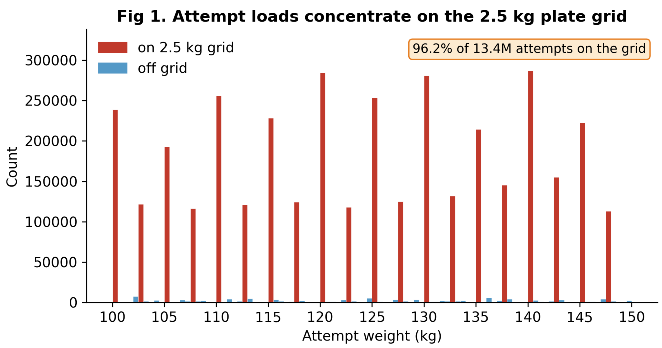

Fig. 1. H1: attempt loads concentrate on the 2.5 kg plate grid.

we restrict to the kg-native (0.5 kg) lattice and model the within-cell position as a mixture that places weight π on the grid point and $1-\pi$ on a uniform background. The GLRT of $\pi = 0$ versus $\pi > 0$ has its null parameter on the boundary of $[0, 1]$, so Wilks' theorem fails: the null distribution of $2 \ln \Lambda$ is the 50:50 mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ of Chernoff and Self–Liang [4], [5]. Our parametric bootstrap recovers exactly this (point mass 0.51 at zero; calibrated 95% cutoff 2.59 versus the naive 3.84). The observed statistic, $2 \ln \Lambda = 3.9 \times 10^7$, exceeds every bootstrap draw ($p < 3.3 \times 10^{-4}$), and a χ^2 goodness-of-fit against a uniform within-cell distribution is overwhelming (5.0×10^7, df $= 4$). The 3.8% off-grid loads are largely a units artifact: of the 2.1% that fall off the 0.5 kg lattice entirely, 82% are round pound loads (e.g., 102.06 kg= 225 lb)—themselves quantized, on a different grid.

B. Bodyweight bunches below class limits (H2)

Bodyweight piles up just below real class limits (Fig. 2). On de-heaped data (exact round/half-kg weigh-ins removed, so the signal is not mere digit preference), the log-ratio of just-below to just-above counts is positive at all seven IPF men's limits and strongest at 83 kg ($+1.92 \pm 0.04$, a 6.8$\times$ asymmetry; Table I, Fig. 3); a non-limit control at 91 kg is flat-to-slightly-negative (-0.21). The formal Cattaneo–Jansson–Ma density-discontinuity test [3] (one random meet per lifter, 2014+) rejects at all seven limits with the density dropping at the cutoff ($t < 0$), surviving both Holm and Benjamini–Hochberg correction; the control shows no excess below (a small *opposite*-direction discontinuity), and no placebo shows the bunching pattern. The effect is robust across equipment (raw $+2.14$, equipped $+1.51$) and tested status (tested $+1.93$), and persists on the clean kg-only subset ($+1.75$). The weakest bunching is at the 120 kg cutoff, which is adjacent to the unbounded 120+ class and therefore carries the weakest incentive to cut.

Robustness and falsification. The result survives four deliberate attacks. (i) *De-heaping*: removing exact round/half-kg weigh-ins leaves the asymmetry intact, so it is manipulation, not digit preference. (ii) *Control*: the non-limit point at 91 kg shows no excess below. (iii) *Placebos*: cutoffs placed between real limits show no bunching-direction discontinuity after correction. (iv) *Subgroups and correction*: the effect holds within raw, equipped, and tested subsets, and the formal density test rejects at all seven limits even under the strictest (Bonferroni) correction.

TABLE I
DE-HEAPED log(BELOW/ABOVE) AT EACH IPF MEN'S LIMIT (ALL ROWS). LAST ROW IS A NON-LIMIT CONTROL.

Limit (kg)	log-ratio	$\pm$	$\times$ asymmetry
59	$+0.99$	0.04	2.7
66	$+1.18$	0.04	3.2
74	$+1.00$	0.03	2.7
83	$\mathbf{+1.92}$	0.04	**6.8**
93	$+1.65$	0.04	5.2
105	$+1.17$	0.04	3.2
120	$+0.74$	0.06	2.1
91 (control)	-0.21	0.03	0.8

C. Allometric scaling differs by sex (H3)

Fitting $\log(\text{Total}) = a + b \log(\text{BW})$ (Fig. 4), the exponent is $b = 0.75$ (CI $[0.74, 0.75]$) for men and 0.51 ($[0.50, 0.51]$) for women, with heteroskedasticity-robust (HC3) standard errors on one row per lifter. A Wald test rejects the isometric $b = 2/3$ for both sexes (men $z = +46.8$, women $z = -64.1$); the one-sided framing is motivated by Neyman–Pearson/UMP logic, exact only under a one-parameter exponential-family model. A pooled Sex$\times$ log(BW) interaction estimates the men-minus-women slope gap at $+0.24$ ($p \ll 10^{-3}$). Crucially, the gap *grows* on the common bodyweight support (0.84 men vs. 0.38 women), so it is not an artifact of the sexes occupying different weight ranges. The Pearson correlation of $\log \text{BW}$ and $\log \text{Total}$ is 0.60 (Fisher-transform CI $[0.60, 0.60]$; Spearman 0.58). The message is not that women are weaker, but that the *estimated scaling* differs and merits a mechanism-level explanation.

D. Strength is predictable from basic variables (H4)

Predicting the total from controllable/basic variables (bodyweight, sex, equipment, age), with every split grouped by lifter and all preprocessing fit inside each training fold, a random forest reaches $R^2 = 0.70$ (RMSE 86 kg) versus 0.61 (RMSE 98 kg) for linear regression (Fig. 5); permutation importance is dominated by sex and bodyweight, and continuous-feature VIFs are ≈ 1 (no collinearity concern). A logistic "made-weight" classifier built from *non-bodyweight* features only—age, tested status, era, equipment, with the Dots score excluded because it is a function of total and bodyweight—reaches AUC $= 0.58$ (PR-AUC 0.88, balanced accuracy 0.54). This is honestly weak: cutting is driven by bodyweight relative to the limit, which the label encodes and the features deliberately exclude, so the modest signal is expected and reassuring rather than disappointing.

E. Supporting analyses

Distribution structure. The apparent two-component shape of raw totals is just sex: by BIC and a bootstrap likelihood-

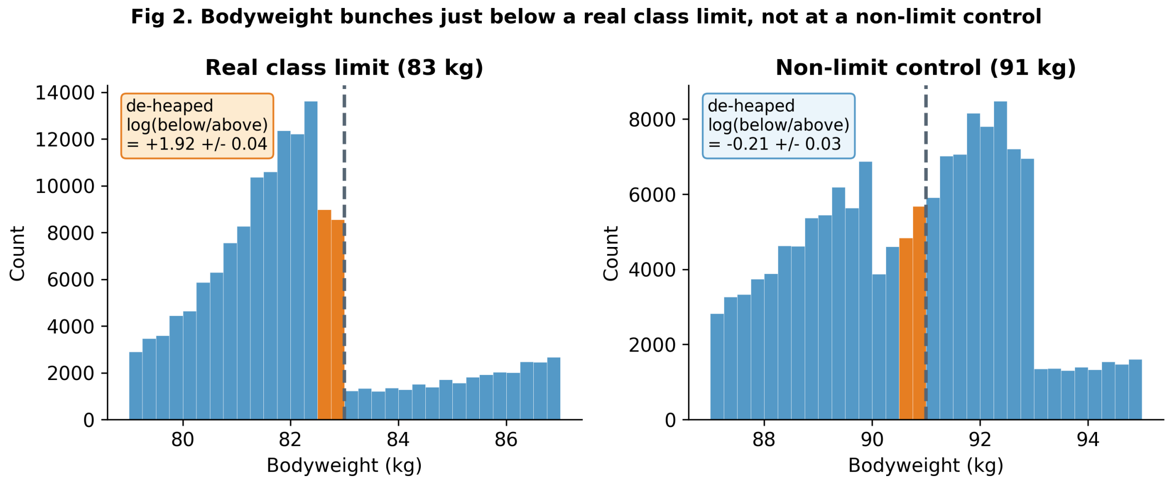

Fig. 2. H2: bunching just below a real limit (83 kg) but not at a non-limit control (91 kg); round-number weigh-ins removed.

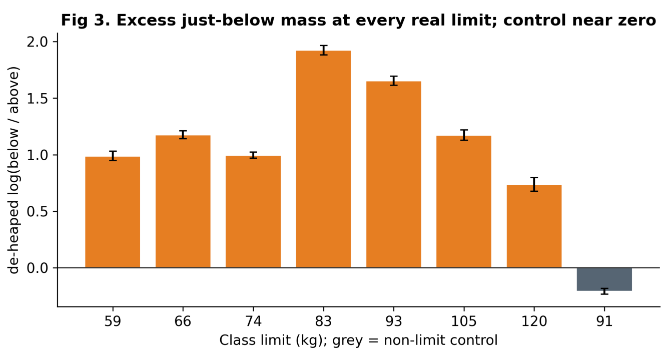

Fig. 3. H2: excess just-below mass at every real limit; control near zero.

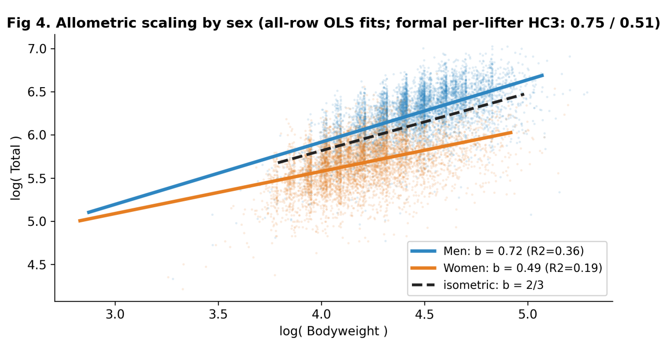

Fig. 4. H3: allometric scaling by sex vs. the isometric $2/3$.

ratio test, raw Total prefers two Gaussian components ($\Delta\text{BIC} = 1.4\times10^3$) but the sex-normalized Dots score prefers one (Fig. 6)—the "mixture" is sex, and normalizing removes it. **Tested vs. untested.** Untested competitions show higher Dots than tested ones (medians 344 vs. 318, Mann–Whitney $p \to 0$, rank-biserial -0.20), consistent with the expected effect of performance-enhancing drugs. **Extreme values.** A GEV fit to equal-size block maxima of recent totals gives shape $\xi = -0.02$ with bootstrap CI $[-0.20, 3.77]$: a population strength *ceiling is not supported*, since the CI includes zero. **Time trend.** Median Dots drifted down over 2000–2024 (Spearman $\rho = -0.54$), consistent with broadening participation rather than declining strength. **Sequential detection.** A Wald sequential probability ratio test on the chronological stream of just-below/above indicators accepts the bunching hypothesis after a stopping time of only $N = 27$ (Fig. 7). **Independence.** After deduplicating to one row per lifter, the association between tested status and cutting is negligible (Cramér's $V = 0.008$), illustrating how pseudo-replication inflates significance.

III. Methods

Data and preprocessing. We use a pinned snapshot of OpenPowerlifting (3,941,811 rows, public domain; the SHA-256 is recorded for exact reproducibility). The unit of analysis is the *attempt* for H1 (the mechanism is within-competitor, so the grid is robust to clustering) and one *row per lifter* for the cross-sectional tests; *de-heaping* drops exact round/half-kg weigh-ins so the bunching signal is not digit preference. For prediction, all cross-validation is grouped by lifter so the same athlete never appears in both train and test.

GLRT and the boundary (H1). With log-likelihood ℓ, the statistic is $2\ln\Lambda = 2(\ell_1 - \ell_0)$. Under regularity $2\ln\Lambda \to \chi^2_r$, but when the null fixes a parameter on the boundary of its space the limit becomes a mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ [4], [5]; we estimate the null distribution directly by simulating data under the fitted null model (parametric bootstrap) and comparing the observed statistic to the simulated quantiles. Concretely, for within-cell position $u \in \{0, \ldots, 4\}$ on the 0.5 kg lattice the model is

$$P(u) = \pi\mathbb{1}[u = 0] + (1 - \pi)\tfrac{1}{5}, \qquad \pi \in [0, 1], \tag{1}$$

and the GLRT contrasts $\pi = 0$ (no grid preference, on the boundary) with the constrained MLE $\hat{\pi} \geq 0$.

Density discontinuity (H2). For a cutoff c the manipulation test contrasts the one-sided densities, $\theta = \ln \hat{f}^-(c) - \ln \hat{f}^+(c)$,

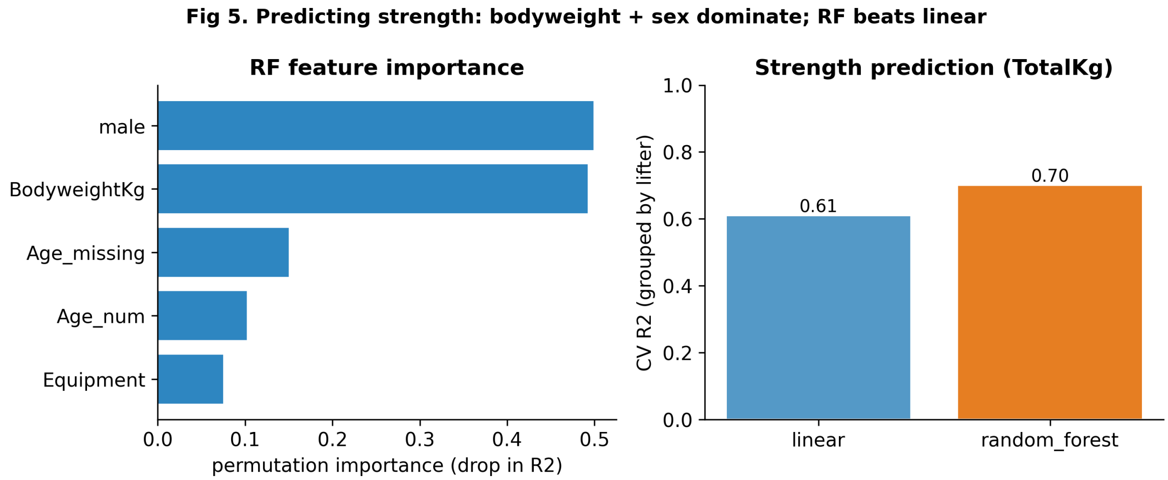

Fig. 5. H4: strength prediction—importance and grouped-CV R^2.

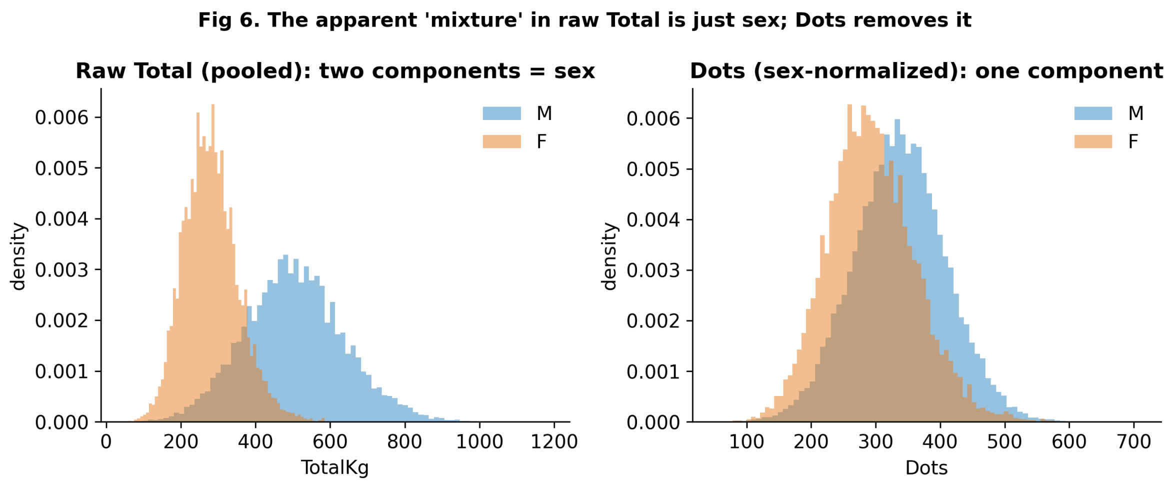

Fig. 6. The apparent “mixture” in raw Total is sex; Dots removes it.

estimated by local polynomials without pre-binning [2], [3]; $\theta < 0$ ($t < 0$) indicates excess mass just below c. We complement the formal test with a de-heaped log count-ratio as the interpretable effect size.

Allometry (H3). OLS on log–log data gives the scaling exponent b; we report HC3/cluster-robust standard errors, a Wald statistic $W = (\hat{b} - 2/3)/\widehat{\mathrm{se}}(\hat{b})$, and a pooled Sex$\times$ log(BW) interaction whose coefficient is exactly the men-minus-women slope difference. Correlations use the Fisher z-transform for confidence intervals.

Learning and the rest of the toolbox. Prediction uses linear and random-forest regression and logistic classification, evaluated by grouped cross-validation (R^2, adjusted R^2, RMSE, MAE, permutation importance, VIF, AUC/PR-AUC). Supporting analyses use a bootstrap mixture-LRT with AIC/BIC, a GEV extreme-value fit, a Wald sequential probability ratio test (with decision boundaries $A = \ln \frac{1-\beta}{\alpha}$, $B = \ln \frac{\beta}{1-\alpha}$, a stopping time, and an expected-sample-size approximation), one- and two-sample nonparametric tests (Wilcoxon signed-rank; Mann–Whitney), an F-test/Kruskal–Wallis omnibus with post-hoc comparisons, and a χ^2 independence test with Cramér's V and standardized residuals. The extreme-value fit uses the GEV distribution

$$G(x) = \exp\Big\{ - \big(1 + \xi \tfrac{x-\mu}{\sigma}\big)^{-1/\xi} \Big\}, \tag{2}$$

where $\xi < 0$ implies a bounded upper tail (a strength ceiling) and $\xi \geq 0$ does not; we estimate ξ from equal-size block maxima and bootstrap its interval. The sequential test's expected sample size under the alternative is approximated by $\mathbb{E}[N \mid H_1] \approx \big[(1-\beta)A + \beta B\big]/\big[p_1 s_1 + (1-p_1)s_0\big]$, with s_0, s_1 the per-observation log-likelihood increments. Table II maps the course toolbox onto the study.

Reproducibility. Every result is produced by a small set of scripts from the pinned snapshot; numbers are written to machine-readable files, figures are exported at 300 DPI, and an

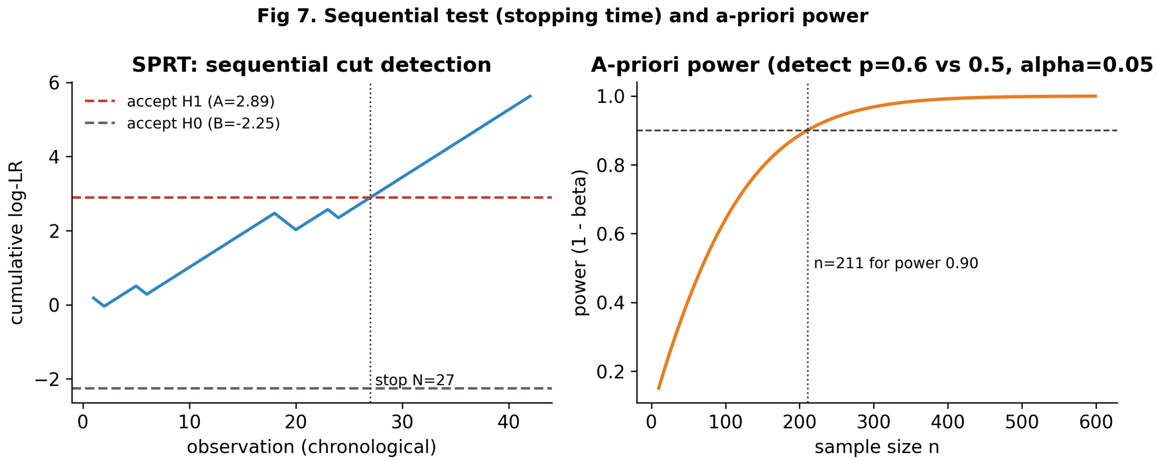

Fig. 7. Sequential test (stopping time $N = 27$) and a-priori power ($n = 211$ for power 0.90).

TABLE II
COVERAGE OF THE COURSE TOOLBOX.

Concept	Where used
GLRT, MLE, χ^2 goodness-of-fit	H1 grid model
Boundary LRT, parametric bootstrap	H1 null calibration
Density discontinuity (McCrary/CJM)	H2 bunching
Multiple testing (Bonf./Holm/Šidák/BH)	H2 limits, H3
OLS, R^2, Wald, interaction term	H3 allometry
One-/two-sided, one-sample tests	H1, H3
Pearson/Spearman, Fisher CI	H3
Regression, random forest, logistic	H4 prediction
Learning metrics, grouped CV, VIF	H4
Mixture LRT, AIC/BIC	structure
Mann–Whitney (two-sample)	tested vs. untested
Wilcoxon signed-rank (paired)	attempt 1 vs. 3
F/ANOVA, Kruskal–Wallis, post-hoc	equipment
χ^2 independence, Cramér's V	tested $\times$ cut
Extreme-value theory (GEV)	strength ceiling
Sequential Wald (SPRT), stopping time	cut detection
Power, Type I/II tradeoff	a-priori power

independent re-derivation from the raw data reproduced every headline value reported here.

IV. Discussion

Conclusions. The two numbers a competitor controls leave measurable, falsification-surviving traces. Attempt loads are quantized to the plate grid; bodyweight bunches specifically below class limits, and not at controls or placebos in the bunching direction. Beyond strategy, the data speak to the limits of strength: scaling with bodyweight is sex-specific and departs from the isometric $2/3$, and basic controllable variables already explain $\sim$70% of the variance in the total. The narrative moves from description to formal test to prediction using one coherent dataset and a single story.

Limitations and null results. This is an observational study: we identify population-level patterns consistent with weight cutting, not the intent of any individual. We report three null results as genuine findings. The unbounded $120+$ class shows the weakest bunching, exactly as incentives predict; the extreme-value analysis does *not* support a strength ceiling (the ξ confidence interval includes zero); and we find no material association between tested status and cutting once pseudo-replication is removed. On the methods side, lifter identity is name-based, the per-lifter meet is sampled at random (we report sensitivity), and the formal bunching test is restricted to the modern kg-only era. Natural extensions—additional federations and eras, a structural bunching model that recovers the share of manipulators, and a multiverse robustness analysis—are left for future work.

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